Bihar Engineering University, Patna
End Semester Examination – 2023

Course: B.Tech. | *Semester-VI* | *Time: 03 Hours*
Code: 101608 | *Subjects: Highway Construction and Management* | *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct option of the following (Any seven questions only):* [2 x 7 = 14]

(a) What is the primary material used in the construction of flexible pavements?
(i) Concrete (ii) Asphalt (iii) Brick (iv) Gravel

(b) Which layer of a flexible pavement is responsible for distributing traffic loads to the lower layers?
(i)Subgrade (ii) Base course (iii) Surface course (iv) Subbase

(c) What is the typical purpose of adding lime or cement to the subgrade in flexible pavement construction?
(i) To improve drainage (ii) To increase strength
(iii) To reduce plasticity (iv) To enhance skid resistance

(d) Which factor primarily influences the design thickness of a flexible pavement?
(i) Traffic volume (ii) Weather conditions
(iii) Subgrade soil type (iv) Asphalt quality

(e) What does the term "PCI" stand for in the context of Pavement Management Systems?
(i) Pavement Condition Index (ii) Pavement Construction Index
(iii) Pavement Cost Index (iv) Pavement Compliance Indicator

(f) Which type of data is typically collected and utilized in a PMS to assess pavement conditions?
(i) Traffic volume data (ii) Weather data
(iii) Surface roughness data (iv) Construction cost data

(g) The Benkelman beam measures the deflection under:
(i) Static load (ii) Dynamic load
(iii) Impact load (iv) Rolling load

(h) The Benkelman beam is specifically designed to measure the deflection of:
(i) Rigid pavements (ii) Flexible pavements
(iii) Composite pavements (iv) Permeable pavements

(i) What is the primary purpose of expansion joints in structures like bridges and buildings?
(i) To prevent vertical movement
(ii) To absorb thermal expansion and contraction
(iii) To increase structural rigidity
(iv) To support additional weight

(j) The longitudinal joints may be
(i) tongue and groove type (ii) butt type
(iii) weakened plane type (iv) All of these

Q.2 (a) Enumerate the steps in the construction of cement concrete pavement. [7]
(b) What is the primary material used for the surface layer in flexible pavement? [7]

Q.3 (a) Explain the importance of soil compaction in earthwork operations. [7]
(b) Describe the significance of slope stability in earthwork projects. [7]

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Q.4 (a) What are the primary functions of the subbase layer in a flexible pavement structure? Describe the key properties required in materials used for the subbase. [7]

(b) Compare and contrast the materials commonly used in subbase construction for flexible pavements, highlighting the advantages and disadvantages of each material in terms of stability, drainage, and load-bearing capacity. [7]

Q.5 (a) Describe the essential characteristics required in materials used for the surface course in flexible pavements. [7]

(b) Compare the contraction joints and construction joints in cement concrete pavements. [7]

Q.6 (a) Discuss the differences between longitudinal and transverse joints in cement concrete pavements. Explain how their design and placement vary based on pavement layout and traffic conditions. [7]

(b) Explain the key components of a Pavement Management System (PMS) and how they interact to facilitate effective pavement management. [7]

Q.7 (a) Discuss the role of life cycle cost analysis (LCCA) in Pavement Management Systems. [7]

(b) Explain the material specification and construction steps for Bituminous surface dressing. [7]

Q.8 (a) Explain the principle and uses of Benkelman Beam test. [7]

(b) What are the various types of failures in flexible pavement. [7]

Q.9 (a) Write short notes on ***any two*** of the following : [3 ½ x2=7]

(i) Draw backs of cement concrete roads
(ii) Expansion joints
(iii) HDM model

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(b) Benkelman Beam deflection studies were carried out on 15 selected points on a stretch of flexible pavement during summer season using a dual wheel load of 4085 kg, 5.6 kg/cm^2 pressure. The deflection values obtained in mm after making the necessary lag corrections are given below. If the present traffic consists of 750 commercial vehicles per day, determine the thickness of bituminous overlay required, if the pavement temperature during the test was 30°C and the correction factor for subsequent increase in subgrade moisture content is 1.3. Assume annual rate of growth of traffic as 7.5%. Adopt IRC guidances. [7]

1.40, 1.32, 1.25, 1.35, 1.48, 1.60, 1.65, 1.55, 1.45, 1.40, 1.36, 1.46, 1.50, 1.52, 1.45 mm

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